Machine Learning for Forecasting Hazardous Air Quality Days in Indian Cities

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MTH5000 Masters Research Project

October 17, 2026

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Motivation

- PM2.5 in Indian cities regularly reaches hazardous levels; sustained exposure raises cardiopulmonary and all-cause mortality
- A forecast only helps if it arrives *before* the hazardous day – same-day warnings are of limited use
- Daily PM2.5 has exploitable structure: day-to-day persistence, a strong annual cycle, dependence on recent conditions
- My own motivation: growing up in India, seasonal smog was ordinary life, not an abstract statistic

Research Problem and Significance

- Existing ML studies of Indian air quality report average error only – not decomposed by the operating point a warning uses
- Good average error can still mean a poor warning system, and the reverse also holds
- The approved proposal assumed a weekly cycle alongside the annual one – tested directly here, not carried over
- **Significance:** if the decision threshold dominates the model, the alarm budget is the more consequential design choice

Objectives and Research Questions

Objectives

1. Evaluate classical, ML and deep learning models 1–3 days ahead, under a leakage-tested protocol
2. Compare forecast-then-threshold against direct classification
3. Quantify skill as a warning system under a varying threshold

Research questions

1. How accurately can PM2.5 be forecast 1–3 days ahead?
2. Threshold a forecast, or classify the exceedance directly?
3. What hit rate / false alarm rate trade-off does each give?

Q3's answer turns out to dominate Q1 and Q2.

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Classical and Machine Learning Methods

- **ARIMA** (Box–Jenkins): differenced series, own past values/errors; order by an AIC grid (Hyndman & Khandakar, 2008); Fourier terms for complex seasonality (De Livera et al., 2011)
- **ML lineage:** ridge (Hoerl & Kennard, 1970); random forest = bagging (Breiman, 1996) + trees (Breiman et al., 2017) + decorrelation (Breiman, 2001); gradient boosting, additive and stagewise (Friedman, 2001)
- **Competition evidence:** flexible learners need not beat ARIMA on short series (M3/M4: Makridakis et al., 2018/2020); an LSTM lost to ARIMA head-to-head (Siami-Namini et al., 2018) – but gradient boosting won M5, where genuine exogenous structure existed (Makridakis et al., 2022)
- **Takeaway:** the deciding factor is exploitable structure, not "ML vs. classical" in the abstract – this project's single series sits closer to M4 than M5
- A single train-test split understates variability (Tashman, 2000; Hewamalage et al., 2023) – the motivation for rolling-origin evaluation

Deep Learning: Why a GRU

- **MLP** (Hornik et al., 1989): no sense of sequence order – needs the same hand-built lag table as the tree models
- **CNN** (LeCun et al., 1998): fixed receptive field, ill-suited to an annual cycle already handed to every model as Fourier terms
- **Transformer / TFT** (Lim et al., 2021): large parameter counts tuned for datasets orders of magnitude bigger than $\sim$3,000 observations – *fitted in full after Dr Tian challenged that premise, and it beats the GRU*
- **Plain RNN:** vanishing/exploding gradients over long sequences (Bengio et al., 1994)
- **LSTM** fixes this with 3 gates and a memory cell – more parameters than this dataset needs (Hochreiter & Schmidhuber, 1997) – *fitted in full on Dr Tian's advice, and it wins too*
- **GRU** (Cho et al., 2014): 2 gates, no memory cell, same gradient-flow property, smaller parameter count – the *a priori* choice. Chapter 4 tests both size arguments, and supports neither

Delhi Air Quality and the Research Gap

- Delhi's emissions: traffic, industry, seasonal stubble burning (Guttikunda et al., 2014; Cusworth et al., 2018; Jethva et al., 2019)
- Prior Delhi PM2.5 ML studies (Masood & Ahmad, 2020; Singh & Srivastava, 2025) score accuracy only – **none** evaluates forecasts as a warning system with an explicit alarm budget
- **Verification framework** (Jolliffe & Stephenson, 2012): match measures to the decision – CSI (Schaefer, 1990); plain accuracy misleads on imbalanced events (He & Garcia, 2009; Saito & Rehmsmeier, 2015)
- **Data quality literature** on negative readings (Jiang et al., 2023) and missingness mirrors the two defects found here
- **The gap:** evaluate against an explicit alarm budget – Chapter 4 shows this changes which model wins

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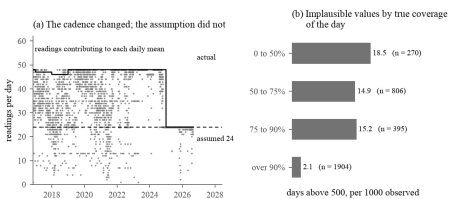
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Data and Two Defects in the Source

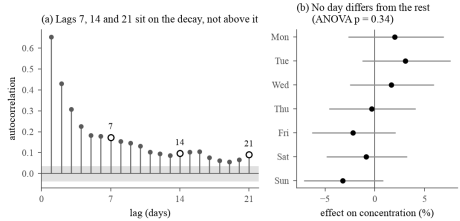
- OpenAQ. Every CPCB station returns only ∼18 months; only location 8118 (AirNow, a US diplomatic post) has the full decade – selected, with the limitation stated plainly
- **Clipped ceiling:** 1990 $\mu g\, m^{-3}$ repeats identically on 3 unrelated dates – an instrument ceiling, not a reading
- **Wrong coverage denominator:** assumed 24 hourly readings, sensor was half-hourly 2016–2024. Recomputed: 1,085 of 3,467 days are below 75% true coverage, vs. 268 with no value at all

Period: 2016–2026
Usable days: 3,093 (86.4%)
Longest gap: 34 days
Days above threshold: 920 (29.7%)



Seasonality: Testing, Not Assuming, a Weekly Cycle

- Proposal assumed a weekly cycle alongside the annual one – tested directly, three independent ways
- **Periodogram:** dominant periods near 358, 179, 61 days – the annual cycle and harmonics, nothing near 7
- **ACF:** lag 7 is 0.173, lag 6 is 0.179 – on the decay curve, not above it
- **Day-of-week:** ANOVA $p = 0.34$; Kruskal-Wallis $p = 0.77$
- **No weekly cycle** – consistent with a diplomatic enclave, not a roadside site. ARIMA carries annual Fourier terms only



Features and Weather Covariates

- One row per origin t, using only data up to day t: lags, rolling mean/SD/max, anomalies, exceedance frequency, calendar and annual Fourier terms, missingness flags
- Missing values are never imputed, except for the three networks, shown next
- **Weather** (`meteostat`): temperature, wind speed, pressure, precipitation, each with a lag and a 3-day rolling mean
- **Third data defect:** the weather archive ends 202 days before the PM2.5 series – a hard cutoff. Precipitation is only 54% complete and excluded from the GRU's weather channels

Classical Benchmark and Machine Learning Regressors

ARIMA(1,1,1) with four pairs of annual Fourier terms, order by AIC (1892.5):

$$(1-\phi_1 B)(1-B)y_t = c + (1+\theta_1 B)\varepsilon_t + \sum_{k=1}^{4}\left[\alpha_k \sin(\tfrac{2\pi kt}{365.25}) + \beta_k \cos(\tfrac{2\pi kt}{365.25})\right]$$

fitted on log(PM2.5); $\hat{\phi}_1 = 0.581$, $\hat{\theta}_1 = -0.987$.

Three ML regressors, each fitted to level and change from origin:

- **Ridge:** closed-form $\hat{\beta} = (X^\top X + \lambda I)^{-1} X^\top y$
- **Random forest:** bagged trees + random feature subset, which decorrelates trees and lowers the variance floor
- **Gradient boosting:** additive, tree per round corrects the residual; accepts missing values natively

The Three Deep Learning Models

The GRU, in full:

$$z_t = \sigma(W_z x_t + U_z h_{t-1} + b_z)$$
$$r_t = \sigma(W_r x_t + U_r h_{t-1} + b_r)$$
$$\tilde{h}_t = \tanh(W_h x_t + U_h(r_t \odot h_{t-1}) + b_h)$$
$$h_t = (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t$$

- **GRU:** 2 gates, no memory cell – 2,329 parameters
- **LSTM:** 3 gates, explicit cell state – 3,097
- **Transformer:** self-attention, $d_{model} = 16$, 2 heads, 1 layer – 3,409, the largest

One protocol, three architectures:

- $H = 24$ hidden units, 30-day window
- 6 channels (10 with weather): log-concentration, missingness flag, Fourier pairs ($K = 2$ vs. ARIMA's $K = 4$)
- Gaps forward-filled, missingness flagged
- Adam, MAE loss, early stopping
- Same channels, window, target, optimiser and seeds – **only the architecture differs**
- **10 seeds** each; the GRU is not bit-reproducible even at a fixed seed, so every result is a range

Self-attention in Place of Recurrence

Every timestep in the 30-day window attends directly to every other one. There is no hidden state propagated step by step – and so no inherent sense of order.

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_{model}}}\right) V$$

$$Z' = \text{LayerNorm}\big(Z + \text{MultiHead}(Z)\big)$$

$$Z'' = \text{LayerNorm}\big(Z' + W_2\,\text{ReLU}(W_1 Z' + b_1) + b_2\big)$$

- A sinusoidal positional encoding is added before the first layer – attention alone cannot tell position
- Two heads on separate subspaces of Q, K, V, so more than one relationship between positions survives
- **Four settings, none independently tuned:** $d_{model} = 16$, 2 heads, 1 encoder layer, and Adam at 0.001 rather than the GRU and LSTM's 0.01 – all inherited from the bounded check. The GRU has three bounded sensitivity checks behind it; the transformer has none yet

Rolling-Origin Evaluation and Warning-System Measures

- **2,303 origins**, Nov 2019–Aug 2026, refit every 90 days (180 for the three networks). At origin t, only data up to day t is used – enforced by construction, verified by a corruption test
- **Accuracy:** rMAE against persistence (1.000 = persistence) and MASE, per horizon
- **Warning system:** hit rate, false alarm rate, precision, CSI, alarms per year – plain accuracy is never reported
- **Diebold-Mariano test** (HLN small-sample correction): each network vs. ARIMA per seed, and ARIMA vs. each ML regressor

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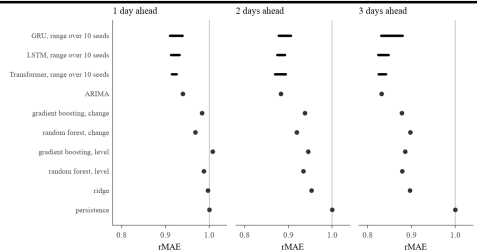
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Forecast Accuracy (RQ1)

Model	$h = 1$	$h = 2$	$h = 3$
Transformer, range over 10 seeds	0.914–0.925	0.870–0.893	0.825–0.842
LSTM, range over 10 seeds	0.913–0.932	0.874–0.892	0.824–0.847
GRU, range over 10 seeds	0.910–0.939	0.878–0.906	0.831–0.879
ARIMA(1,1,1) + Fourier	**0.940**	**0.883**	**0.832**
Best ML regressor (GBM, change)	0.983	0.937	0.878
Persistence	1.000	1.000	1.000



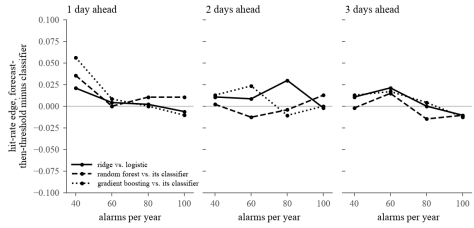
All three networks win clearly at $h = 1$ only; ARIMA wins or ties beyond it. Both architectures added on supervisor review beat the GRU – the LSTM at every horizon, the transformer in 8 of 10 matched seeds. By the DM test at $h = 1$ the transformer is strongest: 7 of 10 seeds significant against ARIMA, to the LSTM's 5 and the GRU's 4. ARIMA still beats every ML *regressor* at every horizon, 15 of 15 DM pairings, 14 at $p < 0.05$.

Does Weather Change the Answer?

- Gradient boosting improves at every horizon (by 8.2 / 2.0 / 2.9 points of rMAE); the random forest worsens 5.3 points at $h = 3$ – no single story for every model
- **GRU + weather:** wins clearly at $h = 1$ (7/10 seeds now significant, up from 4/10); $h = 2$ becomes a genuine 3-way split; $h = 3$ is more often a loss than a win (5 of 10)
- Doubling the seed count (5→10) is what surfaced this – under five seeds alone, $h = 2$ and $h = 3$ had looked like wins
- **Carried through the threshold:** at most 1.2 points of extra hit rate at any budget – a materially more accurate forecaster still does not escape the alarm-budget ceiling

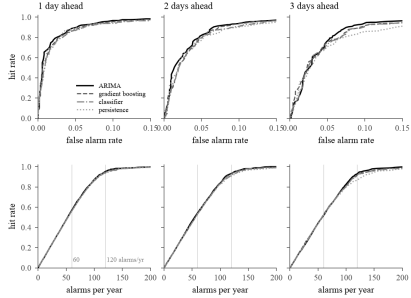
Two Arms of the Same Question (RQ2)

- Forecast-then-threshold vs. direct classification, matched by base algorithm
- 36 comparisons (3 pairs $\times$ 3 horizons $\times$ 4 budgets): **threshold wins 21, classifier 11, 4 ties**
- Median gap 0.011 hit rate, max 0.056 – real but modest
- Edge narrows as budget loosens: 8/9 at budget 40, but the classifier overtakes it (6/9) at budget 100
- Preferred anyway: a forecast can be rethresholded for any future policy; a classifier is welded to its training threshold



The Decision Threshold (RQ3) – the Central Result

Alarms per year	40	60	80	100
Best hit rate, $h = 1$	0.379	0.567	0.735	0.881
Best hit rate, $h = 2$	0.366	0.538	0.700	0.840
Best hit rate, $h = 3$	0.359	0.533	0.685	0.827



Budget 40→100 moves $h = 1$ hit rate 0.379→0.881: **50 points**. Switching the model instead moves it ∼**5 points**. At $h = 3$, budget 40, plain persistence is the best model in the study.

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The Operational Question Dominates the Modelling Question

- Alarm budget moves the hit rate $\sim$50 points; switching the model at fixed budget moves it $\sim$5
- At $h = 3$, loosest budget: climatology alone (0.816) comes within a hundredth of the best fitted model (0.827)
- **Not** an argument against modelling – ARIMA and the GRU both beat persistence, and a better forecast narrows the useful range of thresholds
- The leverage sits with the policy instrument, the alarm budget, not the model behind it. Recommendation: fix the budget with the health authority first; treat the model as a secondary refinement

How Large Is a Difference Between Models?

- **Noise floor:** identical pipeline, two machines, moves tree models up to 0.004; the GRU moves 0.028–0.048 across seeds – almost as much when the *same* seed is simply rerun
- Several table gaps sit at or below that floor (e.g. 0.878 vs. 0.879 at $h = 3$) and cannot be confidently ordered
- **Diebold-Mariano is more conservative still:** only 4/10 GRU seeds reach $p < 0.05$ at $h = 1$ (the noise floor credits 8/10); 0/10 at $h = 2$; 1/10 at $h = 3$
- Those seed counts are themselves small-sample proportions: a 95% Wilson interval on "4 of 10" at $h = 1$ runs 0.17 to 0.69

Limitations I: Scope and Sensitivity

- **Single station, single city:** an AirNow diplomatic-post monitor, not the CPCB network; Delhi only, chosen for data history
- **Cleaning threshold** ($1000\,\mu g\,m^{-3}$) checked directly: restoring capped days barely moves ARIMA ($0.940 \rightarrow 0.935$ at $h = 1$) but hurts the ML suite more (ridge: $0.997 \rightarrow 1.118$) – the central finding *strengthens*
- Tree hyperparameters from a small grid, not an exhaustive search – a lower bound on what those models could achieve

Limitations II: The GRU and Reproducibility

- GRU architecture ($H = 24$, 30-day window) fixed by inspection – a bounded check found a 45-day window scored 2.2 points better at $h = 3$, untested at full ten-seed scale
- **Fourier ablation:** giving the GRU ARIMA's $K = 4$ harmonics helps at every horizon, substantially at $h = 3$ (3.96 points) – the network already gets a seasonal signal, just a coarser one
- GRU training is **not bit-reproducible** at a fixed seed – why the seed count was doubled from 5 to 10
- Weather is observed only, never forecast; boundary layer height is unavailable without a reanalysis product

Limitations III: What the Supervisor Reviews Added

- **Optimiser and learning rate** (bounded check): Adam at 0.01 is at or near best at $h = 1, 2$; at $h = 3$ SGD with momentum beats it by $\sim$2 points (0.839 vs. 0.858). Three bounded checks now agree $h = 3$ is the configuration this project would least defend
- **LSTM and transformer scope:** both got the full ten-seed accuracy evaluation and no further – no weather channels, no threshold sweep for either, so neither appears in the warning-system tables
- **The transformer was promoted out of this list:** it began as a bounded check; Dr Tian challenged the small-data premise ($\sim$4,000 days clears the threshold he had seen cited), so it was given the full treatment – and it beats the GRU in 8 of 10 matched seeds at every horizon
- **KAN-LSTM hybrid** – the one check still bounded. No maintained library, so the layer was built from the published construction and unit-tested alone first. All three seeds trained stably; competitive with the other three

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Summary and Main Conclusions

- Forecast hazardous PM2.5 1–3 days ahead at one Delhi monitor; two real data defects corrected; the weekly-cycle assumption tested and rejected
- Spread across models at a fixed budget: $\sim$5 points; across budgets, model fixed: $\sim$50
- ARIMA beats every ML regressor at every horizon (DM-backed); the three that win are the networks – GRU, LSTM and transformer – each clearly only at $h = 1$, each a range over ten seeds. Both added on review beat the GRU originally chosen
- Weather narrows $h = 1$ further but leaves $h = 2/h = 3$ a tie or a loss; its gain buys little once thresholded
- Forecast-then-threshold beats direct classification 21 of 36; modest, but preferred for rethresholdability
- Trying several methods, not defaulting to ARIMA, surfaced the $h = 1$ network win and made the negative result credible – and two rounds of review then bettered the architecture first chosen

Future Work

1. A genuine weather *forecast*, not the conditions already observed at the forecast origin
2. Boundary layer height via a reanalysis product (e.g. ERA5)
3. Refit the GRU with four Fourier harmonics under the full ten-seed rolling-origin evaluation
4. A second, independent station – ideally CPCB – once its record lengthens
5. Finish the LSTM and the transformer: weather channels and the threshold sweep, the two steps taken for neither
6. Evaluate the KAN-LSTM hybrid in full – the same promotion the transformer was just given

Design the warning system around its alarm budget; treat the forecasting model as a secondary refinement.

Questions?